DUNCO
HOSPITAL MANAGEMENT SYSTEM

Complete System Documentation

& Feature Catalogue

Version 2.4

September 2026

Dunco Web Solutions

https://hmse.duncowebsolutions.co.ke/

https://github.com/dunte1/dunco-hms

Table of Contents

1. Executive Summary

The Dunco Hospital Management System (Dunco HMS) is a comprehensive, web-based healthcare management platform designed for hospitals, clinics, and medical facilities. Built on Laravel 12 with PHP 8.2+, the system provides end-to-end management of hospital operations including patient registration, clinical workflows, billing, pharmacy, laboratory, radiology, human resources, and insurance management.

The system includes SHA (Social Health Authority) integration capabilities through Kenya's Digital Health Authority (DHA) Interoperability Engine, enabling electronic claims processing, eligibility verification, and pre-authorization workflows.

Key Highlights

- 110+ database tables covering all hospital operations
- 21 user roles with granular permission-based access control
- 584+ routes spanning public, authenticated, and API endpoints
- 110+ Eloquent models representing the complete hospital data model
- SHA/EHA integration with OAuth2 authentication and audit logging
- M-Pesa payment integration for mobile money transactions
- RESTful API with Sanctum token-based authentication
- Multi-branch support for hospital networks
- Multi-language support (English, French, Swahili, Arabic)
- Patient registration, profiles, and medical history

2. Product Overview

Technology Stack

Component	Technology
Backend Framework	Laravel 12.x
Programming Language	PHP 8.2+
Frontend	Blade Templates + Alpine.js + Tailwind CSS
Build Tool	Vite 7.0+
Database	MySQL 8.0+ (SQLite for development)
Authentication	Laravel Sanctum (API) + Session-based (Web)
Authorization	Spatie Laravel Permission v6.21
PDF Generation	barryvdh/laravel-dompdf v3.1
Excel Import/Export	maatwebsite/excel v3.1
SMS	Twilio SDK v7.3
Activity Logging	spatie/laravel-activitylog v4.7
Backups	spatie/laravel-backup v9.0
Social Login	Laravel Socialite v5.15
Hosting	Linux Server (cPanel)
Web Server	Apache 2.x

Deployment

- Production URL: https://hmse.duncowebsolutions.co.ke/
- Server: Shared hosting via cPanel (213.139.204.9)
- SSL: Let's Encrypt (active)
- Database: MySQL 8.0 on localhost

3. Key Benefits

- Centralized Patient Management - Complete patient lifecycle from registration through discharge
- Integrated Billing - Automated invoice generation, payment processing, and receipt management
- SHA Compliance - Pre-built integration with Kenya's Social Health Authority
- Real-time Dashboard - Live hospital operations overview with key metrics
- Role-Based Access - Granular permissions for 21 different user roles
- Multi-Branch Support - Manage multiple hospital locations from a single system
- Paperless Operations - Digital records, e-prescriptions, and electronic lab orders
- Mobile Payments - M-Pesa integration for convenient patient payments
- Audit Trail - Complete activity logging for compliance and accountability
- Scalable Architecture - Built on Laravel with queue-based background processing

4. System Modules

The system contains 28 major modules covering all hospital operations:

#	Module	Purpose	Status
1	Administration	System config, user mgmt, roles	Implemented
2	Patient Management	Registration, profiles, history	Implemented
3	Doctor Management	Profiles, departments, schedules	Implemented
4	Appointments	Scheduling, booking, management	Implemented
5	Reception / Queue	Queue management, tokens	Implemented
6	Outpatient (OPD)	Consultations, visits, notes	Implemented
7	Inpatient (IPD)	Admissions, wards, beds	Implemented
8	Nursing	Nurse mgmt, duty roster, wards	Implemented
9	Pharmacy	Medicines, prescriptions	Implemented
10	Laboratory	Tests, requests, results	Implemented
11	Radiology	Imaging tests, requests	Implemented
12	Billing & Finance	Invoices, payments, receipts	Implemented
13	Insurance / SHA	Members, eligibility, claims	Partial
14	Human Resources	Employees, payroll, leave	Implemented
15	Blood Bank	Donors, stock, requests	Implemented
16	Ambulance	Calls, emergency admissions	Implemented
17	Reports	Patient, revenue, financial	Implemented
18	Settings	Config, branches, audit logs	Implemented
19	CMS	Blog, gallery, careers	Implemented
20	Marketing Suite	Posts, campaigns, social	Implemented
21	Patient Portal	Self-service portal	Implemented
22	Telemedicine	Video consultations	Implemented
23	RFID	Tag tracking, scanning	Implemented
24	IoT Bed Monitoring	Sensors, occupancy, alerts	Implemented
25	Biometric	Fingerprint, card scanning	Implemented
26	AI Features	Virtual nurse, predictions	Partial
27	EHR Integration	HL7/FHIR configuration	Partial
28	Finance	Accounts, income, expenses	Implemented

5. User Roles

The system defines 21 user roles with 93 granular permissions across 16 categories:

Role	Description
Super Admin	Full system access to all modules and permissions
Hospital Admin	Hospital operations management
Doctor	Patient care and medical operations
Nurse	Patient care and assistance
Receptionist	Front desk operations, registration
Pharmacist	Medicine management and dispensing
Lab Technician	Laboratory operations
Radiologist	Radiology operations
Accountant	Financial operations
Case Handler	Insurance and case management
Ambulance Operator	Ambulance services
HR Officer	Human resources management
Patient	Limited access to own data via portal
System Auditor	Read-only access across modules
Support Staff	Minimal access
Telemedicine Doctor	Online consultations only
Inventory Manager	Stock management
Procurement Officer	Purchasing operations
IT Support	System maintenance
Marketing Manager	CMS and communication
System AI Bot	Automated operations

6. Insurance / SHA Integration

The system includes SHA (Social Health Authority) integration through Kenya's DHA Health Interoperability Engine (EHA). The ShaService (467 lines) provides production-ready OAuth2 integration.

Current Status

- ShaService - Production-ready OAuth2 integration with DHA/EHA
- EHA Credentials - NOT CONFIGURED in production (fallback to local database)
- InsuranceApiController - SIMULATED responses (hardcoded data)
- Local Workflow - Fully functional for manual SHA claim management
- SHA Member Registry - Database tables created and functional
- ICD-10 Codes - 75 common codes seeded

SHA Integration Capabilities

- OAuth2 client_credentials authentication with token caching
- Patient search and verification against SHA registry
- Eligibility checking, benefits, sub-benefits, interventions
- Consent/visit management with OTP
- Pre-authorization workflow
- Claims submission and tracking
- Full audit logging to insurance_api_logs table
- Graceful fallback to local database when EHA is unavailable

Important Disclaimer

SHA integration is subject to confirmation of the applicable SHA interoperability/API requirements, facility credentials, and approved access from the Digital Health Authority. The system has the technical capability to integrate with SHA via EHA, but live API connectivity has NOT been verified in the production environment.

7. Clinical Workflows

Patient Registration Flow

- Patient arrives at reception
- Registration form completed with personal details
- Auto-generated patient number assigned
- Insurance information recorded (if applicable)
- Patient profile created in the system

Outpatient Consultation

- Patient arrives / called from queue
- Vital signs recorded (temperature, blood pressure, pulse, weight)
- Medical history reviewed
- Clinical examination performed
- Diagnosis recorded (with ICD-10 codes)
- Prescriptions issued
- Lab tests ordered (if needed)
- Follow-up scheduled

Billing Workflow

- Services rendered recorded
- Invoice generated automatically or manually
- Line items added (consultations, procedures, medicines, lab tests)
- Insurance coverage calculated
- Payment received (cash, M-Pesa, card)
- Receipt generated (PDF or thermal)
- Outstanding balance tracked

8. Security

Implemented Controls

Control	Implementation
Authentication	Laravel Sanctum + Session-based
Password Hashing	Bcrypt (Laravel default)
Authorization	Spatie Permission (21 roles, 93 permissions)
CSRF Protection	Laravel CSRF tokens on all forms
XSS Protection	Blade template auto-escaping
SQL Injection	Laravel Eloquent ORM (parameterized queries)
Session Security	Database-backed sessions, HTTPS cookies
API Rate Limiting	60 requests/minute per user/IP
Audit Logging	Spatie ActivityLog for all operations
HTTPS	SSL/TLS encryption via Let's Encrypt
File Upload	Server-side validation on uploads
API Auth	Sanctum token-based
RBAC	Middleware-enforced per route

9. System Architecture

The system follows a standard Laravel MVC architecture with service layers:

- User Layer: Admin, Doctor, Patient, Other Staff
- Web Application: Blade + Alpine.js + Tailwind CSS
- Laravel Application (PHP 8.2): Controllers, Services, Models, Middleware
- Routes: Web (584+) + API (Sanctum-protected)
- Background Processing: Queue Workers, Scheduler, Jobs
- Database: MySQL 8.0 (133 tables)
- External Services: SHA/DHA (EHA), M-Pesa, Twilio SMS, Pusher WebSocket, OpenRouter AI, Zoom

10. Database

The production database contains 133 tables covering all hospital operations. All 110+ migrations have been completed with no pending migrations.

Record Counts (Production)

Table	Count
Users	1
Patients	1
Doctors	0
Appointments	0
Invoices	1
Medicines	0
Lab Requests	0
Prescriptions	0
SHA Members	0
Insurance Claims	0
Employees	1

11. Technical Requirements

Requirement	Minimum	Recommended
PHP Version	8.2+	8.3+
MySQL	8.0+	8.0+
Apache	2.4+	2.4+
Memory	512MB	2GB+
Disk Space	1GB	10GB+
Node.js	18+	20+
Composer	2.x	2.x

12. Current Implementation Status

Component	Status
Core HMS Modules	Fully Implemented
SHA Integration	Partially Implemented
M-Pesa Integration	Configured (sandbox)
SMS (Twilio)	Configured (placeholder)
AI Features	Partially Implemented
EHR (HL7/FHIR)	Partially Implemented
Telemedicine	Implemented
IoT Bed Monitoring	Implemented
RFID Management	Implemented
Biometric	Implemented
Marketing Suite	Implemented
CMS	Implemented
Patient Portal	Implemented

13. Limitations

Item	Priority	Description
SHA EHA Credentials	High	Production EHA credentials not configured
M-Pesa Production	High	Using sandbox URL
Twilio SMS	Medium	Production credentials placeholder
Mail Configuration	Medium	SMTP placeholder values
.env Security	High	File permissions too permissive
Real Data	Medium	Production DB has minimal test data

14. Demo Access

Production Demo URL: https://hmse.duncowebsolutions.co.ke/

Demo credentials should be provisioned separately for prospective clients. Do not use production administrator credentials for demonstrations.

15. Contact Information

Dunco Web Solutions

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Website: https://duncowebsolutions.co.ke/

GitHub: https://github.com/dunte1/dunco-hms

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